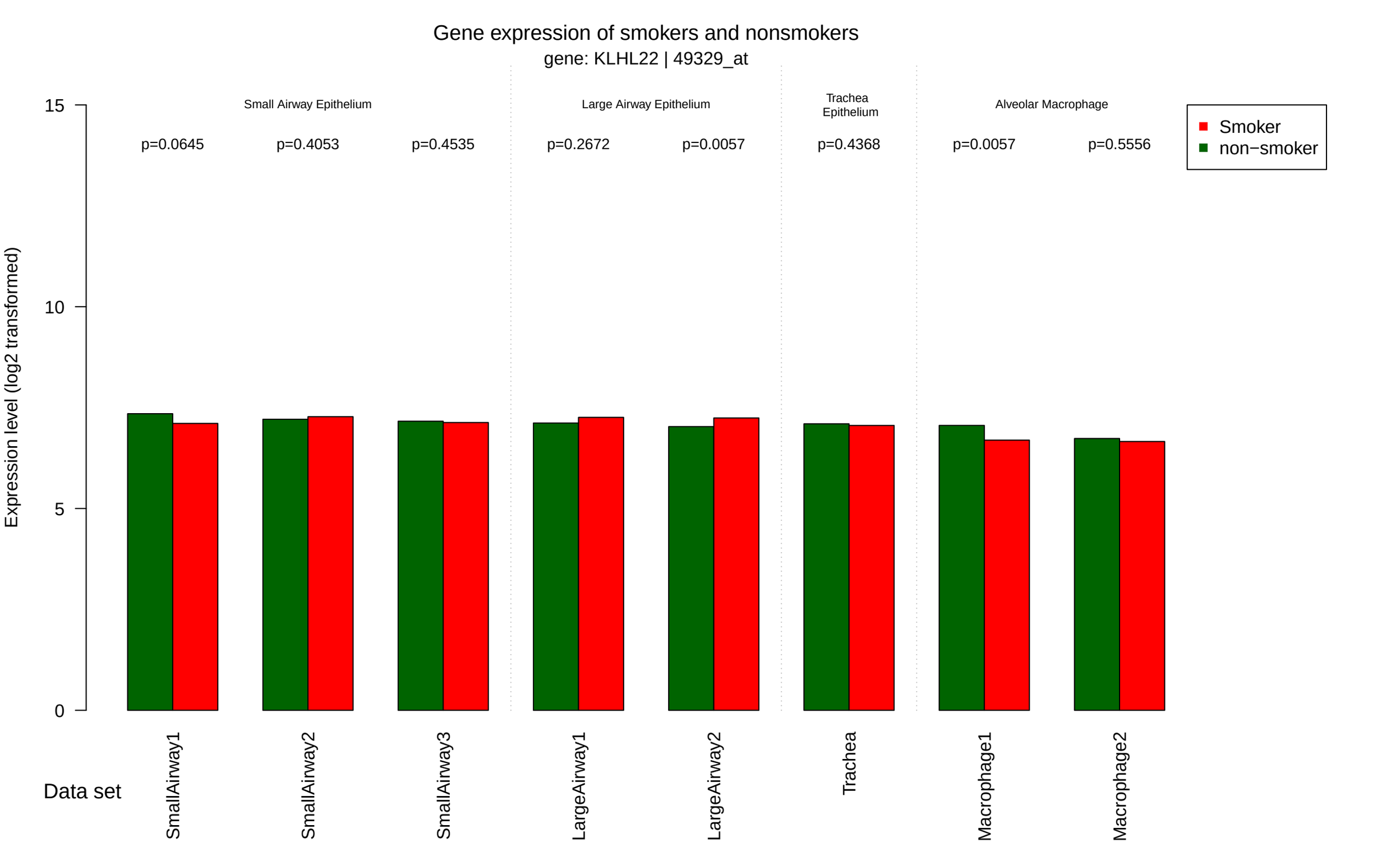
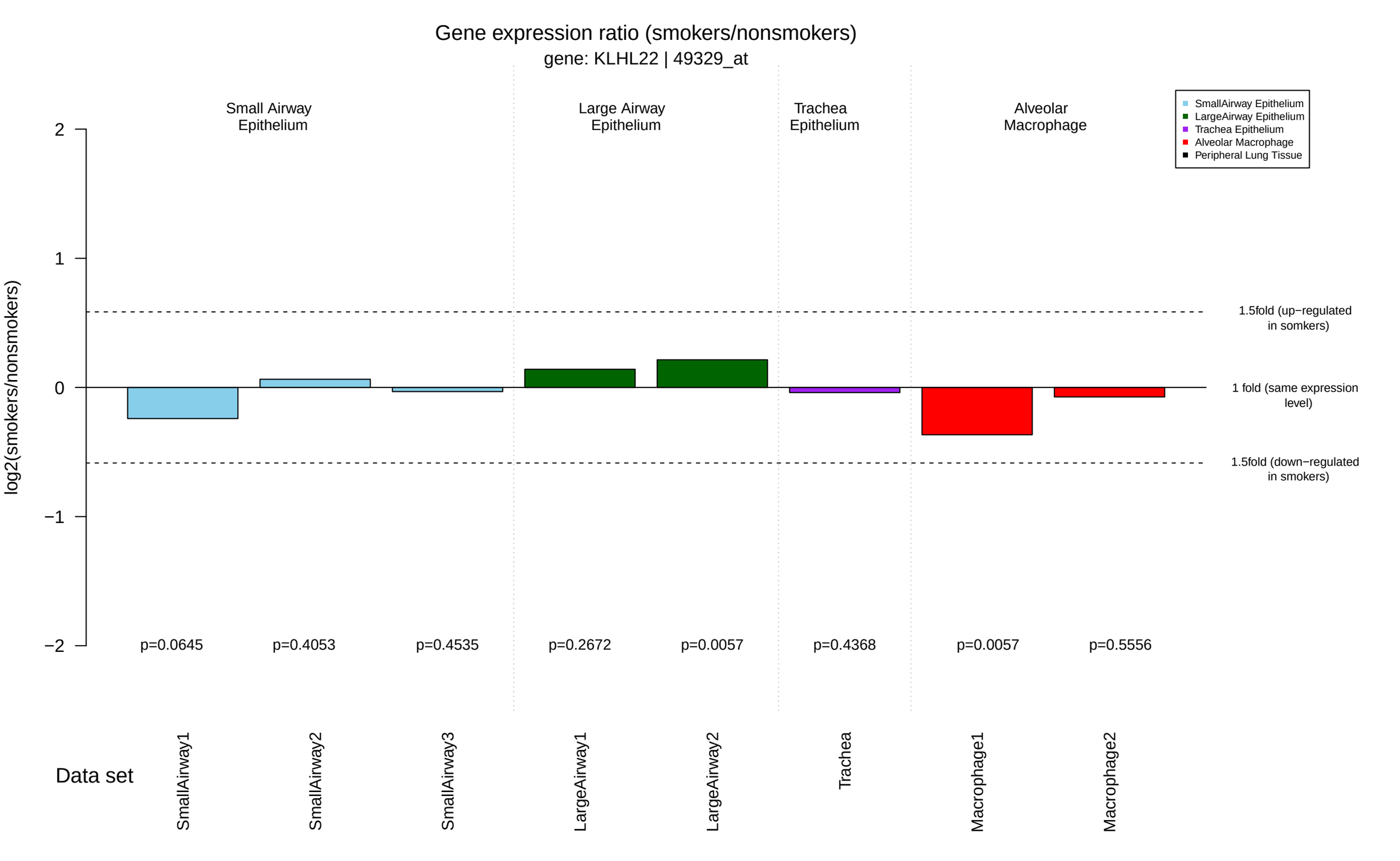
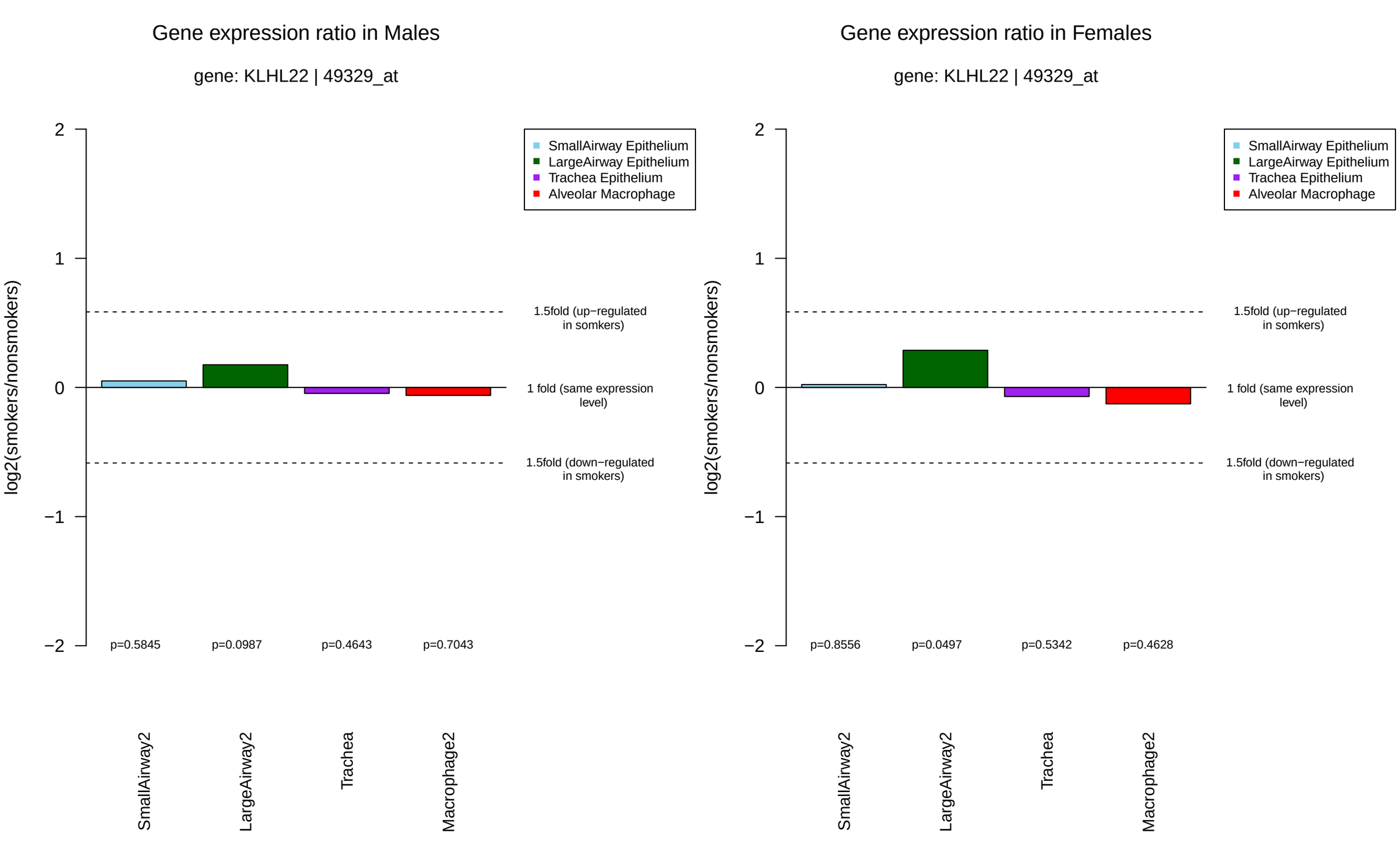
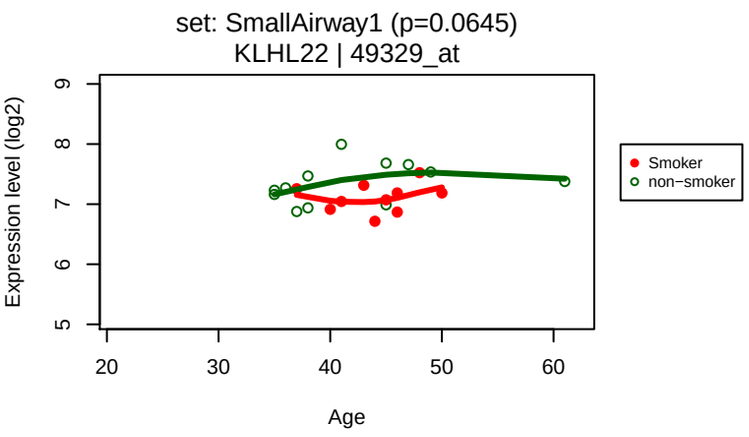
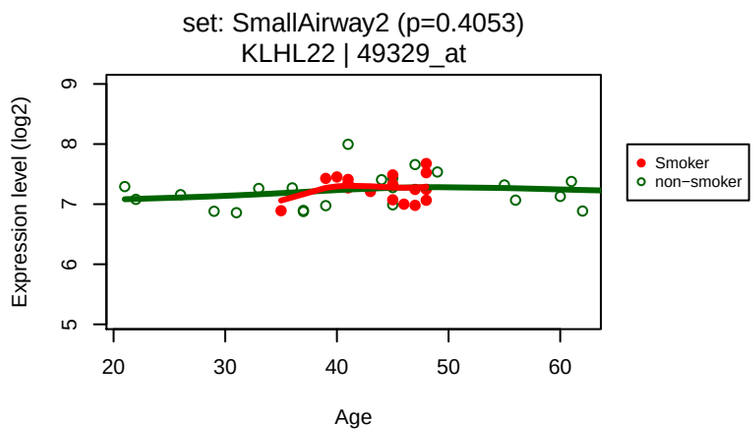
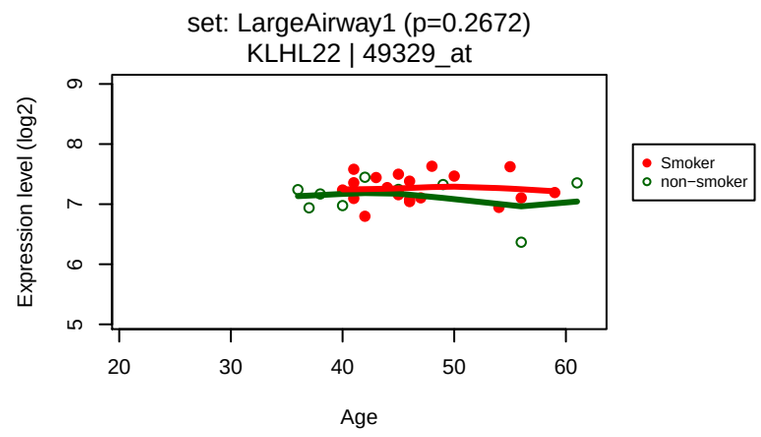

Gene expression of smokers and nonsmokers
gene: KLHL22 | 49329_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0645
p=0.4053
p=0.4535
p=0.2672
p=0.0057
p=0.4368
p=0.0057
p=0.5556
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

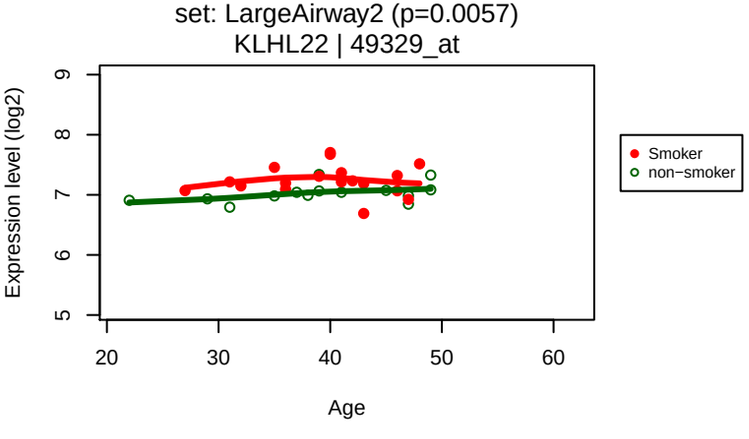

Gene expression ratio (smokers/nonsmokers)
gene: KLHL22 | 49329_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0645
p=0.4053
p=0.4535
p=0.2672
p=0.0057
p=0.4368
p=0.0057
p=0.5556
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

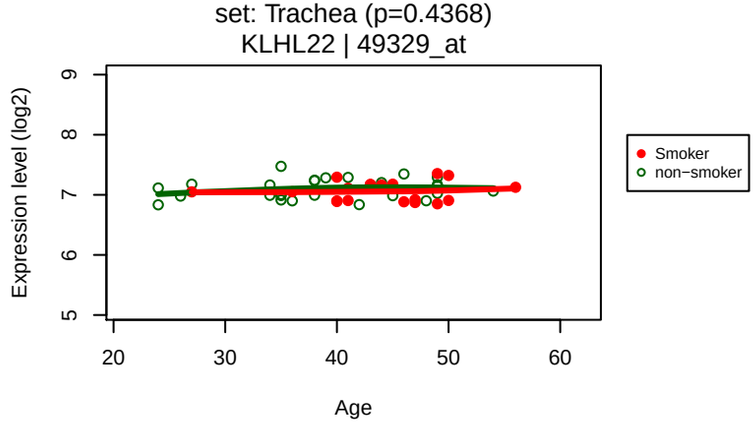

Gene expression ratio in Males
gene: KLHL22 | 49329_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5845
p=0.0987
p=0.4643
p=0.7043
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KLHL22 | 49329_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8556
p=0.0497
p=0.5342
p=0.4628
SmallAirway2
LargeAirway2
Trachea
Macrophage2

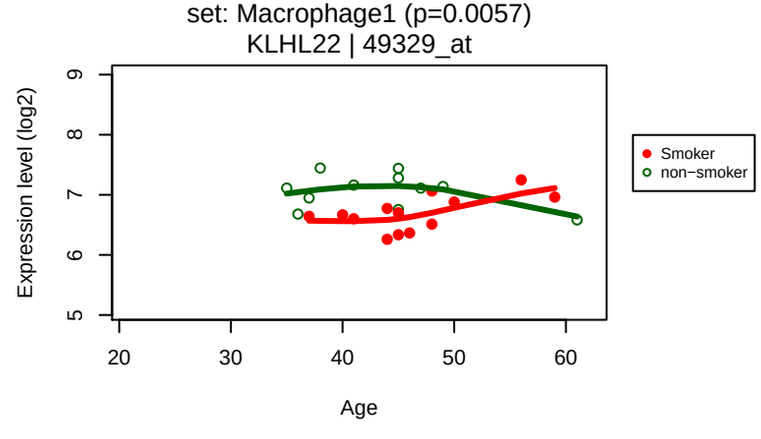

set: SmallAirway1 (p=0.0645)
KLHL22 | 49329_at
Expression level (log2)
Age
Smoker
non−smoker

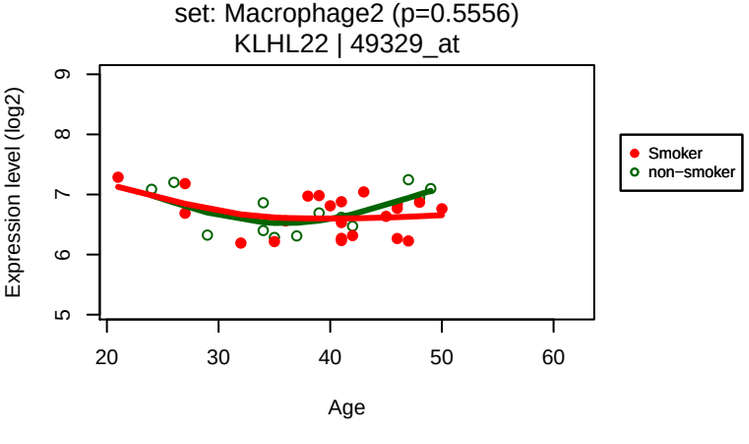

set: SmallAirway2 (p=0.4053)
KLHL22 | 49329_at
Expression level (log2)
Age
Smoker
non−smoker

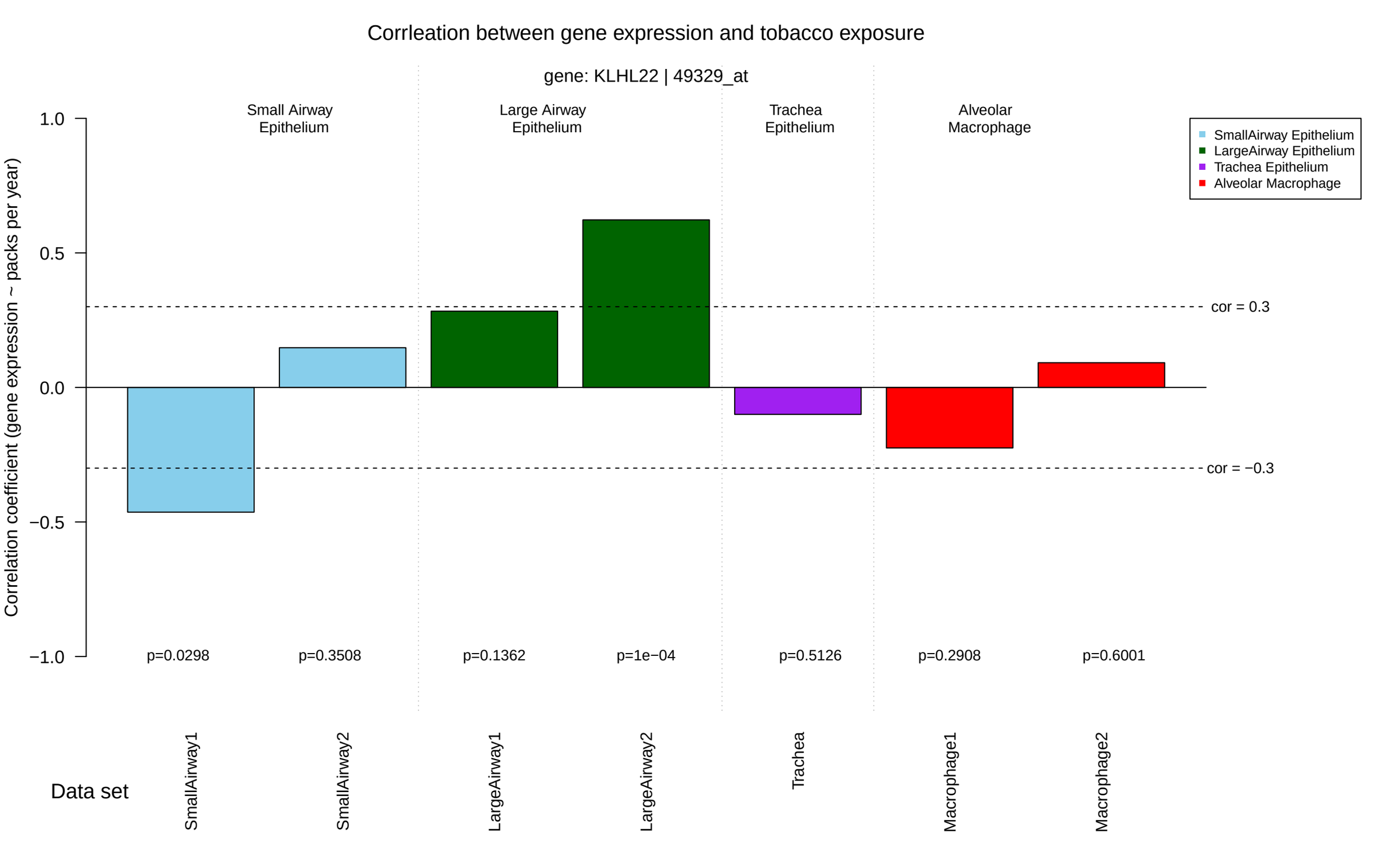

set: LargeAirway1 (p=0.2672)
KLHL22 | 49329_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0057)
KLHL22 | 49329_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4368)
KLHL22 | 49329_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0057)
KLHL22 | 49329_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5556)
KLHL22 | 49329_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KLHL22 | 49329_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0298
p=0.3508
p=0.1362
p=1e−04
p=0.5126
p=0.2908
p=0.6001
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2